

Notes: Handley & Limão (AER, 2017)

Policy Uncertainty, Trade, and Welfare: Theory and Evidence for China and the United States

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1 Big picture

- **Question.** How much did reducing U.S. trade policy uncertainty contribute to China’s post-2001 export boom to the United States, and what were the welfare implications for U.S. consumers?
- **Historical setting.** Before China joined the WTO, the United States renewed China’s MFN status annually. If renewal failed, China would have faced much higher Smoot-Hawley-style column 2 tariffs. WTO accession made MFN treatment permanent and removed the annual threat of a trade war.
- **Main mechanism.** Policy uncertainty discourages exporters from paying sunk costs to enter foreign markets or upgrade export technology. Reducing uncertainty increases export entry, variety, productivity investment, and lowers price indices.
- **Empirical strategy.** Industries differed in their pre-WTO exposure to uncertainty because the gap between U.S. MFN tariffs and column 2 threat tariffs varied strongly across HS-6 products. The paper relates this initial uncertainty to 2000–2005 Chinese export growth, price-index changes, and variety growth.
- **Main empirical result.** Industries with higher initial policy uncertainty experienced faster Chinese export growth and larger price declines after WTO accession.
- **Structural quantification.** A dynamic heterogeneous-firm model with sunk entry and upgrading costs converts reduced-form estimates into general-equilibrium impacts.
- **Magnitude.** The reduction in U.S. trade policy uncertainty accounts for roughly one-third of the growth in Chinese exports to the United States from 2000 to 2005.
- **Welfare.** Removing uncertainty reduced U.S. prices and raised consumer welfare by an amount equivalent to a permanent tariff reduction of about 13 percentage points.
- **Takeaway.** Trade agreements matter not only by lowering current tariffs, but also by making future policy predictable. Binding policy can create large real effects even when applied tariffs barely change.

2 Incremental contribution of the paper

- **From applied tariffs to tariff risk.** Earlier work on trade liberalization focused mainly on actual tariff cuts. This paper shows that a reduction in *uncertainty about future tariffs* can be quantitatively as important as a tariff reduction.
- **A theory-consistent uncertainty measure.** The paper maps observable U.S. MFN and column 2 tariffs into an industry-level measure of the potential profit loss from a policy reversal. This converts a historical institutional feature into an empirically usable treatment intensity.
- **Sunk-cost channel.** The paper connects policy uncertainty to firm entry and export technology investment, rather than only to static trade volumes. This is why price indices and variety respond, not only export values.
- **General-equilibrium quantification.** The paper goes beyond cross-industry regressions by embedding the estimates in a model that accounts for price-index feedback, foreign and domestic entry, and welfare effects.
- **Value of trade agreements.** The paper provides evidence that agreements can be valuable even if they do not immediately change tariffs, because they bind future policy and reduce option-value delays in investment.
- **Distinction from China-shock literature.** The paper studies why Chinese exports rose after WTO accession, while the Autor-Dorn-Hanson China-shock literature focuses on U.S. labor-market consequences. The contribution here is identifying one institutional source of the export boom.

3 Data and measurement

3.1 Trade and price data

- The core empirical unit is an HS-6 industry/product sold by China to the United States.
- The main period is 2000–2005: before and after China’s December 2001 WTO accession, but early enough to focus on the direct post-accession export boom.
- Export growth is measured as log changes in Chinese export values to the United States.
- Price-index changes use variety-level unit values and a CES/Feenstra-type ideal price index that accounts for both price changes of continuing varieties and entry/exit of varieties.
- Variety growth is measured at the HS-10 product-country level within HS-6 industries.

Detailed interpretation: The price-index construction is important because a rise in exports can reflect more units, lower prices, new varieties, or quality/upgrading. The model predicts that reduced uncertainty should change both entry and prices.

3.2 Policy uncertainty measure

Let τ_{1V} be the U.S. MFN tariff factor for industry V , and τ_{2V} be the threatened column 2 tariff factor that would apply if MFN status were revoked. The key observable policy-risk object is the gap between those two tariffs.

A simple way to write the industry-level potential profit loss is:

$$U_V = 1 - \left(\frac{\tau_{1V}}{\tau_{2V}} \right)^\sigma,$$

where σ is the elasticity of substitution. Larger τ_{2V}/τ_{1V} implies larger pre-WTO policy uncertainty.

Detailed interpretation: Industries with large column-2-to-MFN tariff gaps faced large downside risk if U.S. policy reverted to the threat tariff. WTO accession reduced this risk even though actual MFN tariffs changed little.

3.3 Other controls and heterogeneity variables

- Changes in U.S. MFN tariffs control for actual applied tariff changes.
- Changes in transport costs control for non-policy trade-cost changes.
- Sector fixed effects absorb broad HS-section trends.
- Sunk-cost heterogeneity is used to test the mechanism: TPU should matter more where export entry and upgrading require larger sunk investments.
- Taiwan and EU outcomes are used as comparison exercises because Taiwan already had permanent MFN and EU demand should not be directly affected by U.S. China-specific uncertainty.

Detailed interpretation: The design is not simply a before-after comparison of China. It compares products with different pre-WTO exposure to a specific policy-risk reduction, while controlling for contemporaneous tariff and transport-cost changes.

4 Model and empirical specifications

4.1 Exporter problem and policy regimes

The model is a dynamic heterogeneous-firm framework. Firms pay sunk costs to enter the export market and may pay additional sunk costs to upgrade export technology. U.S. policy can be in a low-tariff state τ_1 or a high-threat state τ_2 .

- Higher τ_2 raises downside risk.
- A higher policy-shock arrival probability γ increases the chance of moving into the high-protection state.
- The possibility of a bad tariff state creates an option value of waiting: firms delay entry or upgrading when future policy is uncertain.

Detailed interpretation: Trade policy uncertainty operates like a tax on irreversible investment. It lowers expected returns and raises the entry threshold.

4.2 Baseline export-growth regression

The baseline reduced-form specification relates post-WTO export growth to pre-WTO uncertainty:

$$\Delta \ln X_V = \alpha + \beta_U U_V + \beta_\tau \Delta \ln \tau_V + \beta_d \Delta \ln d_V + \eta_s + \varepsilon_V,$$

where:

- $\Delta \ln X_V$ is 2000–2005 Chinese export value growth in HS-6 industry V ;
- U_V is pre-WTO policy uncertainty;
- $\Delta \ln \tau_V$ is the change in the MFN tariff;
- $\Delta \ln d_V$ is the change in transport cost;
- η_s are sector fixed effects.

Detailed interpretation: A positive β_U means industries with more policy uncertainty before WTO accession grew faster after uncertainty was reduced. This is the central reduced-form prediction.

4.3 Price-index and variety regressions

For the ideal Chinese import price index $P_{V,x}$, the empirical specification is:

$$\Delta \ln P_{V,x} = \alpha + \beta_U^P U_V + \beta_\tau^P \Delta \ln \tau_V + \beta_d^P \Delta \ln d_V + \eta_s + \varepsilon_V.$$

A similar specification is estimated for variety growth.

Detailed interpretation: The model predicts $\beta_U^P < 0$ and a positive effect on variety. Higher pre-WTO uncertainty should be associated with larger price declines and more variety growth after accession.

4.4 Nonlinear structural estimation

The structural model implies nonlinear relationships among uncertainty, tariffs, transport costs, and export growth. The nonlinear least-squares specification estimates key structural objects such as the productivity-upgrading channel and tests restrictions such as:

$$\beta_\tau = \beta_d \frac{\sigma}{\sigma - 1}.$$

Detailed interpretation: The NLLS step matters because linear reduced forms can overstate partial-equilibrium effects. The structural model converts cross-industry coefficients into general-equilibrium export, price, entry, and welfare effects.

5 Identification and empirical design

5.1 Core identifying variation

The identification comes from cross-industry variation in the pre-WTO gap between U.S. column 2 threat tariffs and MFN tariffs. WTO accession sharply reduced the probability that China would face the threat tariff. Products with larger tariff gaps experienced a larger reduction in trade policy uncertainty.

$$\text{Treatment intensity}_V \approx \ln \left(\frac{\tau_{2V}}{\tau_{1V}} \right).$$

Detailed interpretation: The treatment is not the actual tariff cut. It is the reduction in the threat of reverting to much higher tariffs. This is why the paper can identify large effects even though average applied U.S. tariffs toward China barely changed.

5.2 Difference from a standard tariff liberalization design

A standard tariff design would ask whether products with larger applied tariff cuts grew more. Here, MFN tariffs were already in place before accession. WTO accession changed the credibility and permanence of that tariff treatment.

- Applied tariffs: mostly unchanged.
- Threat tariffs: very high and product-specific.
- WTO accession: lowered the probability of the high-threat state.

Detailed interpretation: This makes the setting ideal for measuring policy uncertainty. The policy regime changed while the current tariff schedule remained nearly stable.

5.3 Controls and threats to validity

- **Actual tariff changes.** The regressions control for contemporaneous MFN tariff changes.
- **Transport costs.** The regressions control for changes in transport costs.
- **Sector shocks.** Sector fixed effects absorb broad HS-section demand and supply shifts.
- **Non-tariff barriers.** Robustness checks control for non-tariff barrier changes.
- **Processing trade.** Robustness checks drop processing-trade-intensive industries.
- **China-specific supply shocks.** Taiwan and EU comparison exercises check whether the results are driven by China-specific productivity changes unrelated to U.S. policy uncertainty.

Detailed interpretation: The main threat is that high-uncertainty industries may have been on different growth trajectories for other reasons. The paper addresses this with controls, placebo/comparison samples, sunk-cost heterogeneity, and price/variety mechanisms.

5.4 Sunk-cost heterogeneity as mechanism validation

The model predicts that policy uncertainty matters most where entry or upgrading requires large sunk costs. Therefore the paper interacts uncertainty with high-sunk-cost indicators.

$$\Delta \ln X_V = \alpha + \beta_H U_V \times HighSunkCost_V + \beta_L U_V \times LowSunkCost_V + \dots$$

Detailed interpretation: If the mechanism is option value of waiting, effects should be larger in high-sunk-cost sectors. The tables show precisely this pattern for exports, prices, and variety.

5.5 Price and variety as additional validation

If reduced TPU raises exports through entry and upgrading, then it should also affect import prices and variety. The paper therefore estimates price-index and variety regressions.

- More entry raises variety and lowers the CES price index.
- Upgrading or productivity improvements lower prices of continuing varieties.
- The price effect is not merely a quantity response; it is a mechanism check.

Detailed interpretation: The price and variety evidence is critical because it links the export response to firm investment margins, not just to reallocation of existing trade flows.

5.6 Quantification logic

The reduced-form estimates show that TPU matters. The structural model then asks how much of the aggregate China export boom and U.S. welfare gain can be attributed to this channel.

1. Estimate industry-level responses to TPU reductions.
2. Use the model to adjust for price-index feedback and general equilibrium.
3. Aggregate across industries using expenditure weights.
4. Convert the effects into ad valorem equivalent tariff changes and welfare impacts.

Detailed interpretation: This is why the paper can make statements such as “one-third of the export boom” and “13 percentage-point tariff equivalent.” These are not directly read from a regression coefficient; they come from the estimated model.

6 Tables and figures

6.1 Figure 1: Chinese Import Penetration in the United States; Figure 2: Sector-Level Chinese Export and Price Index Growth versus Initial Uncertainty

Read: Figure 1 compares actual Chinese import penetration with a counterfactual path in which U.S. TPU is reintroduced after WTO accession. Figure 2 provides sector-level motivation: sectors with larger initial uncertainty have faster export growth and larger price declines.

Detailed interpretation: Figure 1 is the headline quantitative claim: the observed import penetration increase is far above the counterfactual with continuing uncertainty. Figure 2 shows that the cross-sectional variation needed for identification is visible even before running the full regression model. The price panel is especially important because a pure demand boom would raise quantities but not necessarily lower Chinese price indices. These two figures belong together because they connect aggregate facts with cross-industry identification. Figure 1 states the aggregate puzzle; Figure 2 shows the policy-uncertainty gradient that the paper exploits.

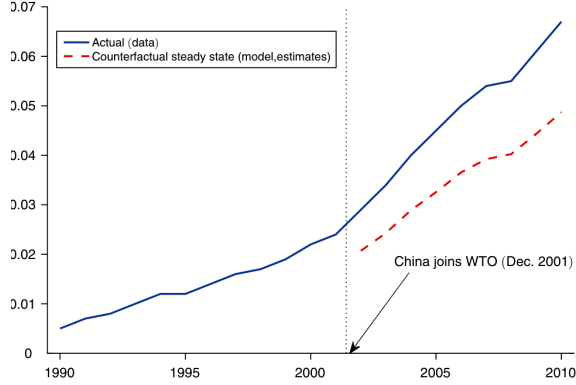


FIGURE 1. CHINESE IMPORT PENETRATION IN UNITED STATES: ACTUAL VERSUS COUNTERFACTUAL UNDER POLICY UNCERTAINTY

(a) Figure 1. Actual import penetration rises much faster than the counterfactual with reintroduced TPU.

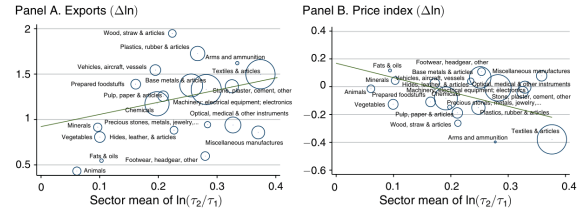


FIGURE 2. SECTOR-LEVEL CHINESE EXPORT AND PRICE INDEX GROWTH VERSUS INITIAL UNCERTAINTY

Notes: Simple means within sector of export and price index change versus means of initial uncertainty measured by $\ln(\tau_{2V}/\tau_{1V})$, where τ_{2V} and τ_{1V} are the column 2 and MFN tariff factors in 2000 in HS-6 industry V . Circles are proportional to the number of observations used as weights in the linear fit represented.

enough to affect the importer's aggregate outcomes. Otherwise, TPU has no significant effect. (b) Figure 2. Higher initial tariff uncertainty predicts faster export growth and lower price-index growth.

6.2 Table 1: Summary Statistics by Pre-WTO Policy Uncertainty; Table 2: Chinese Export Growth Baseline Estimates

Read: Table 1 shows that high-uncertainty industries have higher export growth, more variety growth, larger threat tariffs, and a larger column-2-to-MFN tariff ratio. Table 2 shows that pre-WTO uncertainty positively predicts 2000–2005 Chinese export growth after controlling for tariff changes, transport costs, and sector fixed effects.

Detailed interpretation: Table 1 verifies that the treatment intensity is meaningful: high-uncertainty products differ strongly in tariff risk. Table 2 is the main reduced-form export result. The coefficient on uncertainty is consistently positive, while tariff and transport-cost changes have the predicted negative signs. This supports the view that uncertainty reductions, not applied tariff cuts, are central. These two tables form the empirical starting point. Table 1 establishes the cross-sectional variation; Table 2 estimates its export-growth effect.

TABLE 1—SUMMARY STATISTICS BY PRE-WTO POLICY UNCERTAINTY

	Uncertainty		Total
	Low	High	
Chinese export value growth ($\Delta \ln$, 2005 – 2000) ^a	1.16 [1.77]	1.35 [1.62]	1.28 [1.68]
Chinese export price index growth ($\Delta \ln$, 2005 – 2000) ^a	-0.07 [0.70]	-0.14 [0.69]	-0.11 [0.69]
Chinese export variety growth ($\Delta \ln$, 2005 – 2000) ^a	0.27 [0.46]	0.35 [0.41]	0.32 [0.43]
MFN tariff ($\ln$), 2000	0.03 [0.04]	0.04 [0.05]	0.04 [0.05]
Column 2 tariff ($\ln$), 2000	0.16 [0.10]	0.39 [0.12]	0.31 [0.16]
Ratio of column 2 to MFN tariff	1.14 [0.09]	1.42 [0.14]	1.33 [0.18]
Uncertainty (pre-WTO) ^b	0.30 [0.18]	0.64 [0.09]	0.52 [0.20]
Change in MFN tariff ($\Delta \ln$, 2005 – 2000)	-0.002 [0.01]	-0.004 [0.01]	-0.003 [0.01]
Change in transport costs ($\Delta \ln$, 2005 – 2000)	-0.01 [0.10]	-0.002 [0.08]	-0.005 [0.09]
Observations ^c	1,124	2,087	3,211

Notes: Simple means with standard deviations in brackets. Low: subsample of industries in the bottom tercile of pre-WTO uncertainty (ranked by τ_2/τ_1); High refers to the rest of the sample. Total includes the full sample used in baseline Table 2.

^aTest of mean difference across samples significant at least at the 5 percent level.

^bPotential profit loss measure if MFN revoked pre-WTO: $1 - (\tau_1/\tau_2)^2$, using $\sigma = 3$.

^cTotal observations for price index change: 2,579 HS-6 industries used in baseline Table 4. Total observations for variety growth (number of traded HS-10 varieties with an HS-6 industry): 3,211.

(a) Table 1. High-uncertainty industries have larger tariff gaps and faster subsequent export growth.

TABLE 2—CHINESE EXPORT GROWTH ($US, \Delta \ln$): BASELINE ESTIMATES, 2000–2005

	(1)	(2)	(3)	(4)
Uncertainty pre-WTO	0.743 [0.154]	0.791 [0.150]	0.716 [0.186]	0.734 [0.184]
Change in tariff ($\Delta \ln$)	-9.967 [4.478]	-4.34 [0.676]	-7.356 [5.060]	-4.25 [0.677]
Change in transport costs ($\Delta \ln$)	-2.806 [0.455]	-2.893 [0.450]	-2.795 [0.456]	-2.833 [0.451]
Constant	0.851 [0.0853]	0.843 [0.0850]		
Observations	3,211	3,211	3,211	3,211
R^2	0.03	NA	0.05	NA
Sector fixed effects	No	No	Yes	Yes
Restriction p -value (F -test)	0.204	1	0.536	1

Notes: Robust standard errors in brackets. Predicted sign of coefficient in brackets under variable. Uncertainty measure uses US MFN and column 2 Tariffs to construct profit loss measure at $\sigma = 3$. All specifications employ OLS and 2 and 4 impose theoretical constraint on tariffs and transport cost coefficients: $b = b_1/\sigma/(1 - \sigma)$. Sectors defined by the 21 HS sections.

(b) Table 2. Baseline regressions show a robust positive uncertainty coefficient.

6.3 Figure 3: Chinese Export and Price Index Growth in High versus Low Uncertainty Industries; Figure 4: Chinese Export and Price Index Growth versus Initial Policy Uncertainty

Read: Figure 3 compares distributions of export and price-index growth for high- and low-uncertainty industries. Figure 4 shows nonparametric and semiparametric relationships between growth outcomes and the uncertainty measure.

Detailed interpretation: The density plots show that the high-uncertainty distribution shifts toward higher export growth and lower price-index growth. The local polynomial and semiparametric panels show that the relationship is not driven only by a linear functional-form assumption. The monotone export response and downward price response are exactly the joint signature predicted by the sunk-cost uncertainty mechanism. These figures validate the shape of the treatment effect before the tables impose parametric regressions. They also justify the later nonlinear structural specification.

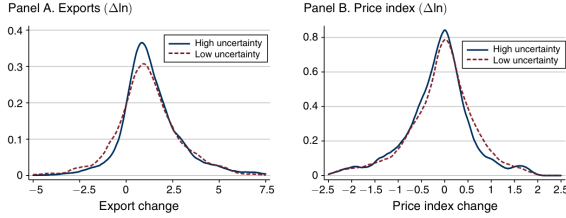


FIGURE 3. CHINESE EXPORT AND PRICE INDEX GROWTH IN HIGH VERSUS LOW UNCERTAINTY INDUSTRIES

Notes: Epanechnikov kernel density estimates. High uncertainty defined as the top two terciles of ratio of US column 2 to MFN tariff in HS-6 industry V . Low uncertainty is the bottom tercile. Equality of distributions rejected

(a) Figure 3. High-uncertainty sectors have more mass at high export growth and lower price growth.

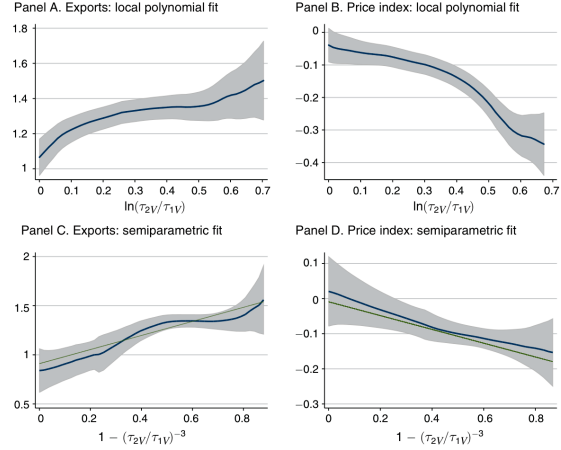


FIGURE 4. CHINESE EXPORT AND PRICE INDEX GROWTH ($\Delta \ln$) VERSUS INITIAL POLICY UNCERTAINTY

Notes: Panels A and B are local polynomial fits for $f(\cdot)$ where τ_{2V}/τ_{1V} is the ratio of column 2 to MFN tariff

(b) Figure 4. Nonparametric and semiparametric fits show positive export and negative price effects.

6.4 Table 3: US Import and Chinese Export Growth, Sunk Cost Heterogeneity; Table 4: Chinese Price Index and Variety Growth

Read: Table 3 tests whether the export effect is stronger in high-sunk-cost sectors and whether comparison samples support the China-to-U.S. interpretation. Table 4 shows that uncertainty reductions lower Chinese price indices and raise variety growth.

Detailed interpretation: Table 3 is a mechanism and placebo table. The uncertainty effect is concentrated where sunk costs are high, consistent with option-value delays. Taiwan and EU comparison exercises help rule out simple global sectoral shocks. Table 4 extends the outcome set: TPU reductions affect not only export values but also prices and varieties, which are the margins the model emphasizes. Together, the tables show that the export boom is not just a quantity fact. It is accompanied by price and variety responses consistent with entry and upgrading.

TABLE 3—US IMPORT AND CHINESE EXPORT GROWTH: SUNK COST HETEROGENEITY AND UNOBSERVED SHOCKS, 2000–2005

	China		Taiwan		Pooled	
	Low (1)	High (2)	Low (3)	High (4)	Low (5)	High (6)
US imports from: Sunk cost sample:						
<i>Panel A. US import growth ($\Delta \ln$): robustness to unobserved HS-6 import demand shocks in high versus low sunk cost industries</i>						
Uncertainty pre-WTO (US)	−0.611 [0.393]	1.105 [0.280]	−0.593 [0.414]	0.0486 [0.297]		
Uncertainty pre-WTO (US) $\times$ 1(China)					0.0663 [0.517]	1.026 [0.383]
Change in tariff ($\Delta \ln$)	−9.59 [13.30]	−12.0 [7.107]	−6.699 [11.91]	−19.3 [6.484]		
Change in transport costs ($\Delta \ln$)	−2.28 [1.200]	−4.51 [0.743]	−3.21 [0.892]	−5.18 [0.724]	−1.12 [0.880]	−4.67 [0.797]
Observations	759	1,519	759	1,519	1,518	3,038
R^2	0.08	0.09	0.13	0.11	0.32	0.43
Sector $\times$ country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
HS-6 fixed effects	No	No	No	No	Yes	Yes
<i>Panel B. Chinese export growth ($\Delta \ln$): robustness to unobserved HS-6 export supply shocks in high versus low sunk cost industries</i>						
Uncertainty pre-WTO (US)	0.0363 [0.313]	0.816 [0.263]	0.492 [0.302]	−0.336 [0.241]		
Uncertainty pre-WTO (US) $\times$ 1(China)					−0.444 [0.409]	1.144 [0.315]
Change in tariff ($\Delta \ln$)	−2.659 [11.49]	−7.703 [5.805]	5.845 [6.075]	−12.43 [3.166]	−1.15 [5.881]	−9.563 [3.876]
Observations	975	1,945	975	1,945	1,950	3,890
R^2	0.047	0.037	0.066	0.05	0.03	0.04
Sector $\times$ country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
HS-6 fixed effects	No	No	No	No	Yes	Yes

Notes: Robust standard errors in brackets clustered on HS-6 industry in columns 5 and 6. Uncertainty pre-WTO is defined as in the baseline US sample. Total sample in columns 1 and 2 reduced relative to Table 2 because sunk cost estimates not available for all HS-6 industries. The change in the US MFN tariff does not vary across nonpreference

(a) Table 3. Effects are stronger in high-sunk-cost sectors and comparison samples support the identification.

TABLE 4—CHINESE PRICE INDEX AND VARIETY GROWTH (US, $\Delta \ln$), 2000–2005

	Price index				Variety			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Uncertainty pre-WTO	−0.284 [0.0689]	−0.217 [0.0828]	−0.418 [0.173]	−0.483 [0.223]	0.0736 [0.0309]	0.0606 [0.0380]	0.243 [0.0722]	0.200 [0.0907]
MFN tariff ($\Delta \ln$)	5.137 [1.608]	0.560 [1.682]	9.839 [4.313]	6.829 [4.497]	−5.170 [0.957]	−4.066 [1.024]	−3.196 [1.179]	−2.792 [1.185]
Transport cost ($\Delta \ln$)	−0.435 [0.254]	−0.460 [0.248]	−0.477 [0.626]	−0.465 [0.620]	−0.227 [0.0909]	−0.200 [0.0897]	−0.514 [0.165]	−0.504 [0.160]
Observations	2,579	2,579	907	907	2,579	2,579	1,051	1,051
Industry sample	HS6	HS6	HS6	HS4	HS6	HS6	HS6, uncensored	HS6, uncensored
R^2	0.02	0.06	0.02	0.06	0.03	0.06	0.03	0.07
Sector fixed effects	No	Yes	No	Yes	No	Yes	No	Yes
Uncertainty price impact ($\Delta \ln$) ^a								
Average price	−0.15	−0.11	−0.22	−0.25				
Aggregate price	−0.17	−0.13	−0.25	−0.28				

Notes: Robust standard errors in brackets. All specifications employ OLS. Constant or sector fixed effects included but not reported. The dependent variable is the $\ln$ change in the ideal price index of Chinese varieties sold in the United States between 2000 and 2005 calculated at the industry level; see online Appendix C for details. Sample: we use the subset of industries where value and quantity data are available and price changes are defined for at least one HS-10 variety in 2000 and 2005 for the industry (HS-6 in columns 1–2 or HS-4 in columns 3–4). The HS-6 sample trims outliers in the 2.5 percent tails of the dependent variable. Columns 7 and 8 exclude censored HS-6 industries, i.e., those where all their HS-10 categories had positive trade in both periods (these observations contain

(b) Table 4. Higher initial TPU predicts lower Chinese price-index growth and greater variety growth.

6.5 Table 5: Chinese Price Index and Variety Growth, Sunk Cost Heterogeneity; Table 6: Chinese Export Growth, Nonlinear and Linear Estimates

Read: Table 5 shows that price and variety effects are stronger in high-sunk-cost industries. Table 6 compares nonlinear least squares and OLS estimates and tests theoretical restrictions from the model.

Detailed interpretation: Table 5 reinforces the sunk-cost channel for the price and variety margins. The lack of comparable effects in low-sunk-cost sectors is hard to reconcile with a generic China productivity story. Table 6 is the bridge between reduced-form evidence and quantification: it estimates the nonlinear model and verifies that the structural restrictions are not rejected in the data. These tables are the transition from mechanism evidence to model-based welfare calculations. They show that the structural model is not just imposed after the fact; it is disciplined by observed export responses.

TABLE 5—CHINESE PRICE INDEX AND VARIETY GROWTH: SUNK COST HETEROGENEITY
(US, $\Delta \ln$), 2000–2005

	Price index		Variety	
	(1)	(2)	(3)	(4)
Uncertainty pre-WTO				
× High sunk cost indicator	−0.449 [0.0816]	−0.320 [0.0923]	0.096 [0.0423]	0.082 [0.0485]
× Low sunk cost indicator $[-0]$	−0.009 [0.118]	−0.062 [0.130]	0.019 [0.0396]	0.030 [0.0471]
MFN tariff ($\Delta \ln$)	4.429 [1.610]	0.332 [1.687]	−4.710 [0.950]	−3.788 [1.012]
Transport cost ($\Delta \ln$)	−0.446 [0.253]	−0.465 [0.248]	−0.217 [0.0896]	−0.196 [0.0888]
High sunk cost indicator	0.165 [0.0793]	0.103 [0.0810]	0.029 [0.0324]	0.029 [0.0345]
Constant	−0.047 [0.0658]		0.055 [0.0230]	
Observations	2,579	2,579	2,579	2,579
R^2	0.02	0.06	0.04	0.07
Sector fixed effects	No	Yes	No	Yes

Notes: Robust standard errors in brackets. All specifications employ OLS. Dependent variables and controls described in notes to Table 4. High sunk cost indicator is 1 for industries in top two

(a) Table 5. TPU affects price and variety mainly where sunk costs are high.

TABLE 6—CHINESE EXPORT GROWTH: NONLINEAR AND LINEAR ESTIMATES (US, $\Delta \ln$), 2000–2005

	NLLS (1)	OLS (2)	NLLS (3)	OLS (4)
Uncertainty (pre-WTO)	0.903 [0.324]	0.686 [0.150]	0.712 [0.345]	0.567 [0.185]
MFN tariff ($\Delta \ln$)	−6.678 [1.26]	−6.464 [1.266]	−6.446 [1.268]	−6.34 [1.270]
Transport cost ($\Delta \ln$)	−4.452 [0.84]	−4.309 [0.844]	−4.298 [0.845]	−4.227 [0.847]
Constant	1.598 [0.109]	0.877 [0.0845]		
Observations	3,043	3,043	3,043	3,043
R^2	0.02	.	0.04	.
Sector fixed effects	No	No	Yes	Yes
Number of coefficients estimated	3	3	23	23
Restriction test $\sigma = 3$ (p -value)	0.11	NA	0.98	NA
Restriction test $b_\tau = b_d \times \sigma/(\sigma - 1)$ (p -value)	0.335	0.373	0.72	0.818

Notes: Standard errors in brackets. Columns 1 and 3 use nonlinear least squares and columns 2 and 4 use ordinary least squares. Predicted sign of coefficient in brackets under variable. Sample: all specifications exclude transport cost outliers, as measured by changes in costs that exceed the top or bottom quartile by more than three times the value of the interquartile range. Uncertainty measure uses US MFN (τ_1) and column 2 tariffs (τ_2) to construct the profit loss measure. This is approximated by $1 - (\tau_1/\tau_2)^\sigma$ under OLS. For NLLS we do not approximate and use instead the general function $\ln(1 + b_\tau \times (\tau_1/\tau_2)^\sigma)$ where b_τ is estimated as described in the text. The four specifications in the columns restrict $\sigma = 3$. We test this by relaxing the restriction in two additional NLLS specifications; we report the p -values in the second to last line at which we can't reject the restriction. We also impose the restriction that $b_\tau = b_d \times \sigma/(\sigma - 1)$. The last row reports p -values from the test of this restriction.

parameters have the following structural interpretation: $b_d = -k$
(b) Table 6. NLLS estimates and restriction tests connect the reduced form to the model.

6.6 Figure 5: Domestic and Foreign Variety Price Indices; Figure 6: Aggregate Price, Sales, Entry

Read: Figure 5 shows how introducing U.S. TPU increases the foreign variety price index and lowers the domestic variety price index. Figure 6 traces model-implied changes in aggregate prices, export sales, export entry, domestic sales, and domestic entry as policy uncertainty, threat tariffs, or applied tariffs vary.

Detailed interpretation: Figure 5 illustrates the core general-equilibrium price channel: less foreign entry raises foreign prices, while domestic entry and price adjustments partly offset the aggregate effect. Figure 6 shows that TPU has nonlinear effects. Foreign export entry and sales fall sharply as uncertainty rises, while domestic entry and sales can rise because foreign competition weakens. This is why partial-equilibrium coefficients do not mechanically equal welfare effects. These two figures translate the empirical estimates into the model's equilibrium economics. They explain why the welfare consequences depend on both foreign and domestic entry.

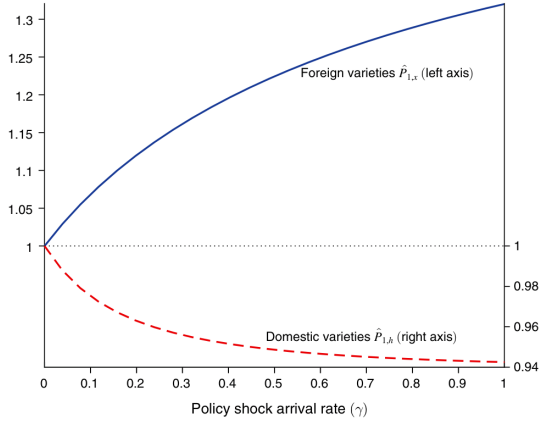


FIGURE 5. DOMESTIC AND FOREIGN VARIETY PRICE INDICES (Counterfactual Introduction of US TPU on All Partners)

: General equilibrium solution of the model for each variable's growth relative to deterministic baseline if the United States introduces TPU in 2005. Solutions computed from estimated and other parameters in Table

(a) Figure 5. TPU raises foreign variety prices and lowers domestic variety prices through entry responses.

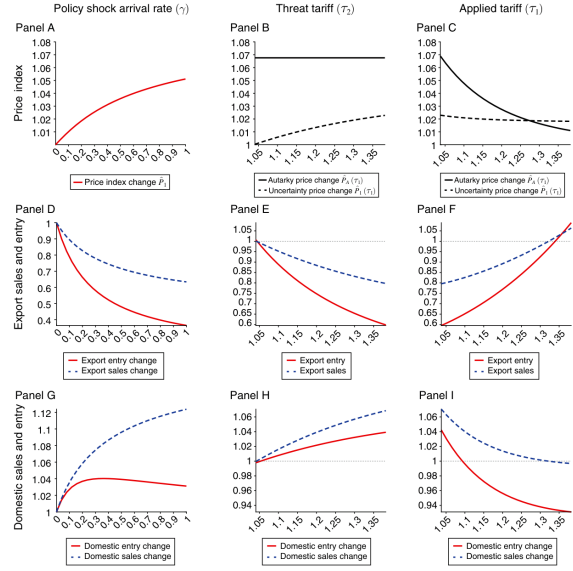


FIGURE 6. AGGREGATE PRICE, SALES, ENTRY (Counterfactual Introduction of US TPU on All Partners)

Notes: Model solution for a variable's growth relative to deterministic baseline if the United States introduces TPU in 2005 using parameters in Table A11. For panels B, E, and H, the threat tariff (τ_2) ranges from the mean observed MFN tariff of 1.041 to the column 2 tariffs of 1.38, both in 2000. For panels C, F, and I, the applied tariff (τ_1) ranges from the simple mean of the observed MFN tariff in 2005 of 1.038 to the column 2 tariffs in 2000 of 1.38. Aggregate

(b) Figure 6. Counterfactual TPU changes prices, sales, and entry for both foreign and domestic firms.

6.7 Table 7: Impact of TPU Reduction on Chinese Export Value and Price; Table 8: Impact on Chinese and US Outcomes and Tariff Ad Valorem Equivalent

Read: Table 7 decomposes the effect of the TPU reduction on Chinese export values and prices, contrasting partial-equilibrium estimates, general-equilibrium estimates, and data equivalents. Table 8 converts model effects into aggregate changes and tariff-equivalent magnitudes.

Detailed interpretation: Table 7 shows that accounting for general equilibrium lowers the estimated export effect relative to the partial effect but still leaves a large 28–32 log point increase in exports. The price effect is also large and negative. Table 8 is the policy valuation table: TPU reduction is equivalent to a 5-point tariff reduction for Chinese exports, a 13-point tariff reduction for the U.S. price index and welfare, and a large reduction in export entry costs. These are the paper's final quantitative takeaways. They turn the historical WTO accession episode into welfare-relevant numbers and show why trade agreements can matter even without large contemporaneous tariff cuts.

TABLE 7—IMPACT OF TPU REDUCTION ON CHINESE EXPORT VALUE AND PRICE
($US, 100 \times \Delta \ln$), 2000–2005

	Policy uncertainty reduction estimates					
	Partial effect		General equilibrium effect		Data equivalent	
	Average	Aggregate	Average	Aggregate	Average	Aggregate
Export value	36	40	28	32	112	73.0
Export price	−15.2	−17.2	−14.8	−17	−26	−13

Notes: The first four columns are the model estimates for the growth in Chinese export value (or price) due to TPU reduction. The general equilibrium employs the coefficients from NLLS export estimates in column 1 of Table 6 and uses the model to incorporate the price index effects. The partial effect estimates ignore the price index effect and use OLS estimates for exports (column 2 of Table 6) and prices (column 1 of Table 4). The data equivalents for exports use the observed Chinese export growth (average or aggregate) and subtract nominal growth in the aggregate US expenditure on manufactures to account for nominal effects and aggregate expenditure shocks, which are held constant in the model prediction. The price growth data equivalent uses the average from Table 1 (or its aggregate using ideal weights) and subtracts the corresponding average (or aggregate) for non-Chinese varieties (used in Table A9) to account for common nominal shocks. The difference in the price data equivalent is mostly driven by the Chinese differential average and aggregate price growth.

(a) Table 7. General-equilibrium estimates attribute around one-third of the Chinese export boom to TPU reduction.

TABLE 8—IMPACT ON CHINESE AND US OUTCOMES OF TPU REDUCTION AND CORRESPONDING TARIFF AD VALOREM EQUIVALENT ($100 \times \Delta \ln$), 2000–2005

	AVE (tariff equivalent)	Aggregate change
Chinese (real) export value [−]	5.0	32.4
Chinese export entry and investment [−]	9.4	61.2
Chinese export price index [−]	6.1	−16.9
US price index (manufacturing) [−]	13	−0.5
US consumer welfare [−]	13	−0.4
US (real) domestic sales (manufacturing) [−]	5.9	−1.3
US domestic employment (manufacturing) [−]	7.3	−1.2
US firm entry and investment (manufacturing) [−]	1.8	−0.4

Notes: Quantification uses NLLS estimates (column 1, Table 6). AVE (ad valorem equivalent) refers to the equivalent tariff reduction that would induce the same change in outcome x in the deterministic model with no uncertainty as the TPU reduction does. See text for details of calculation. Equivalents for outcome x in any period after uncertainty changes but applied policies remain unchanged. Reported values reflect differences in steady-state outcomes between policy states. [−/+] denotes sign of derivative of column variables with regard to tariff.

(b) Table 8. TPU reduction has large tariff-equivalent effects on U.S. prices and welfare.

7 Short summary

- The paper studies China’s WTO accession as a reduction in U.S. trade policy uncertainty rather than only a trade liberalization.
- Pre-WTO China faced annual MFN renewal risk; a failure would have triggered much higher column 2 tariffs.
- Industries differed in initial uncertainty because the gap between MFN and column 2 tariffs varied by product.
- Higher initial uncertainty predicts faster post-WTO Chinese export growth, larger price declines, and more variety growth.
- The effects are stronger in high-sunk-cost industries, consistent with the option-value channel.
- Comparison exercises using Taiwan, the EU, sector fixed effects, non-tariff barriers, processing trade, and alternative samples support the interpretation.
- A dynamic heterogeneous-firm model maps reduced-form estimates into aggregate and welfare effects.
- The reduction in TPU accounts for over one-third of the growth in Chinese exports to the United States from 2000 to 2005.
- The U.S. welfare gain is equivalent to a roughly 13 percentage-point permanent tariff reduction.
- The central lesson is that trade agreements can be valuable by removing uncertainty even when applied tariffs barely change.

8 Conclusion

8.1 Core conclusion

Handley and Limão show that policy uncertainty is a first-order determinant of trade, prices, and welfare. China’s WTO accession mattered because it converted temporary MFN access into a credible permanent commitment. This reduction in the threat of a trade war stimulated entry and upgrading by Chinese exporters, expanded varieties, lowered U.S. prices, and increased real income.

8.2 Economic intuition

When firms must pay sunk costs to enter export markets, uncertainty about future market access makes investment less attractive. Even if current tariffs are low, the possibility of a future tariff reversal acts like a risk on irreversible investments. WTO accession reduced this risk. The result is more entry, more upgrading, and lower prices.

8.3 Why the contribution is important

The paper changes how we should think about the value of trade agreements. Their value is not only in tariff reductions written into schedules. Agreements also bind future policy, reduce uncertainty, and unlock private investment. This is especially important in settings where current policies are favorable but politically fragile.

8.4 Policy implication

A country can generate large gains by credibly committing not to raise future trade barriers. Conversely, increasing uncertainty about future market access can reduce trade and welfare even before any tariff is actually imposed. This insight applies to WTO accession, trade wars, Brexit-style negotiations, and any setting in which firms face uncertainty over future policy regimes.

8.5 Limitations and caution

The design is strongest for China-U.S. manufacturing trade in the early 2000s, where policy uncertainty was observable through the MFN-versus-column-2 tariff gap. In other settings, policy uncertainty may be harder to measure, and the firm-level sunk-cost channel may differ by sector. The quantification also relies on structural assumptions about elasticities, entry, and aggregation.

8.6 Takeaway

The paper's enduring message is that predictability is a trade policy instrument. Removing tariff uncertainty can create export growth, price reductions, and welfare gains comparable to large permanent tariff cuts. Trade agreements are valuable not merely because they lower policy barriers, but because they make policy less risky.